

Light and Photosynthesis

Student worksheet | Biology 2e, Chapter 8 | 45-60 minutes

Name: _____ Class: _____ Date: _____

1. Predict and plan

How does light level affect the time it takes leaf disks to float?

Observe oxygen accumulation in infiltrated leaf disks. Compare three light conditions, repeat each treatment, and use flotation time as an indirect measure of net photosynthetic activity.

Prediction and scientific reasoning:

Variables and controls, or sampling plan:

Keep consistent: 10 same-sized disks, same leaf type, 0.2% sodium bicarbonate solution with a trace of detergent, solution depth, and temperature. Only light level changes.

Select your route

Part	Virtual / Classroom	Equipment or conditions
Dark control		
Low light		
Bright light		

Use the same questions for either route. Always label the source of each dataset.

Light and Photosynthesis

Student worksheet / observations | Biology 2e, Chapter 8 | 45-60 minutes

2. Collect observations

Record floating disks (of 10). Repeat each treatment with fresh material for three independent trials.

Treatment / trial	0 min	2 min	4 min	6 min	8 min	10 min	12 min	14 min	16 min	18 min	20 min
Dark control / 1											
Dark control / 2											
Dark control / 3											
Low light / 1											
Low light / 2											
Low light / 3											
Bright light / 1											
Bright light / 2											
Bright light / 3											

Source (virtual/classroom) for each treatment: _____

Trial summaries

Treatment	Trial 1	Trial 2	Trial 3	Mean
Dark control				
Low light				
Bright light				

Metric: ET50 in minutes. Use the first crossing of five disks; interpolate between readings. If not reached by 20 min, report that limit. Calculate a mean only when all three trials reach ET50.

Observations and graph

Graph time on the x-axis and the recorded measurement on the y-axis on separate graph paper, or use the app. Label treatments, units, and data sources. Record unusual observations below.

Light and Photosynthesis

Student worksheet / explanation | Biology 2e, Chapter 8 | 45-60 minutes

3. Explain the evidence

ET50 is the interpolated time when 5 of 10 disks float. Lower ET50 suggests faster net oxygen accumulation under matched conditions. If five disks never float, report "not reached within 20 minutes"; do not treat that as a zero rate.

1. Which light condition had the shortest ET50? Support your claim with values from at least two treatments and discuss variation between trials.

2. Why are the dark control and matched leaf preparation necessary?

3. Explain how oxygen accumulation makes disks float and why this is an indirect measure of net activity, not a direct measurement of sugar production.

4. Identify one limitation or source of uncertainty and propose an improved investigation.

Before sharing: check units, source labels, calculations, and evidence. The app saves on this device only; download or print your report.

Cut & sort: matter and energy

Optional introduction or extension | 10-15 minutes | Biology 2e, chapters 7 and 8.

Use the optional Build the equation activity on the lab prediction page, or arrange the printed cards on chart paper. The online activity groups identical molecules; the paper activity uses all 19 molecule cards.

Print cutout pages in color, single-sided, at 100% scale. Cut along dashed lines. Use scissors safely. Group work: one person arranges, one counts atoms, and one explains; then rotate roles.

Model photosynthesis. Sort all the molecule cards into reactants and products. Add plus signs and the arrow, and decide where light and chlorophyll belong.

Count carbon, hydrogen, and oxygen atoms separately on each side. Explain how your arrangement shows conservation of matter and the transfer of energy. Energy symbols are annotations, not part of this simplified atom inventory.

This is an overall net equation, not a sequence of reaction stages. The glucose diagram shows an atom inventory, not a molecular structure.

Name: _____ Class: _____ Date: _____

My equation and symbol placement:

Atom counts: Reactants C ____ H ____ O ____ | Products C ____ H ____ O ____

Explain where energy enters, is stored, or leaves this process:

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

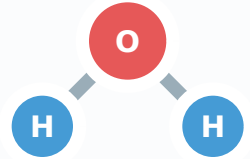
Build a balanced equation. Place light and chlorophyll where they help.



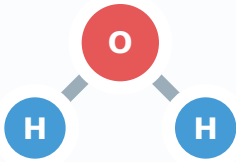
Oxygen



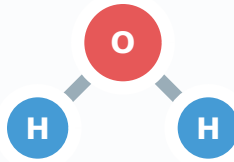
Carbon dioxide



Water



Water



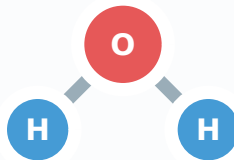
Water



Oxygen



Carbon dioxide



Water



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 1/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

+



Combine with

O₂



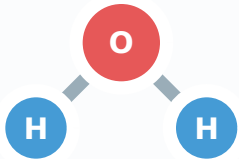
Oxygen

CO₂



Carbon dioxide

H₂O



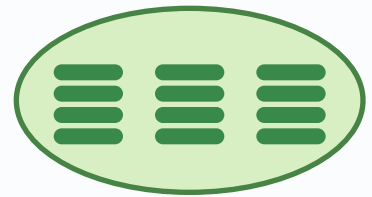
Water

+



Combine with

CHLOROPHYLL



Green pigment in chloroplasts
Simplified chloroplast diagram
Light-absorbing pigment

C₆H₁₂O₆



Atom inventory (not structure)

Glucose

O₂



Oxygen

LIGHT ENERGY



Sunlight

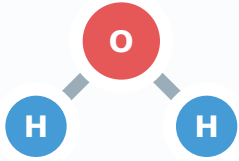
Atom colors: C = gray • O = red • H = blue

Student cutouts 2/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.



Water



Reaction arrow



Oxygen



Oxygen



Carbon dioxide



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 3/3 • Print in color, single-sided, at 100%.